

Zitong Lan

ztlan@seas.upenn.edu | zitonglan.github.io

EDUCATION

- **University of Pennsylvania, Philadelphia** Aug. 2023 - Now.
Ph.D. in Electrical and System Engineering GPA: 3.96 Advisor: Mingmin Zhao
- **University of Pennsylvania, Philadelphia** Aug. 2023 - Aug. 2026
M.S. in Electrical and System Engineering
- **Southeast University, Nanjing, China** Sep. 2019 - Jun. 2023
B.E. in Electrical Engineering with honor GPA: 3.96 Rank: 4/160

WORK EXPERIENCE

- **Research Intern, Microsoft, Redmond** May. 2026 - Aug. 2026
Topic: Enabling Streaming Memory for Full Duplex Speech Language Model
Mentor: Dung Tran, Vasily Zadorozhnyy, Kazuhito Koishida

RESEARCH INTEREST

My research aims to develop multimodal foundation models that can understand, and generate different modalities including text, audio and speech, enabling AI agents to perceive and interact with the physical world through sound.

PUBLICATIONS

- **Zitong Lan**, Mutian Tong, Jiatao Gu, Mingmin Zhao, “Enabling Immersive Audio-Visual Experience from Any Video”. (Arxiv 2026)([Project Site](#))
- **Zitong Lan**, Yiwei Tang, Yuhan Wang, Haowen Lai, Yido Hao, Mingmin Zhao, “Building Audio-Visual Digital Twins with Smartphones”. (Mobisys 2026)([Project Site](#))
- **Zitong Lan**, Yiduo Hao, Mingmin Zhao, “SmartDJ: declarative audio editing with audio language model”. (ICLR 2026, NeurIPS 2025 GenProCC **Oral**)([Project Site](#))
- **Zitong Lan**, Yiduo Hao, Mingmin Zhao, “Resounding the acoustic field through reciprocity learning”. (NeurIPS 2025)([Project Site](#))
- **Zitong Lan**, Chenhao Zheng, Zhiwei Zheng, Mingmin Zhao. “Acoustic Volume Rendering for Neural Impulse Response Fields”. (NeurIPS 2024 **Spotlight**)([Project Site](#))
- Yiduo Hao*, Yiwei Tang*, Jiayang Li*, **Zitong Lan***, Haowen Lai, Romit Roy Choudhury, and Mingmin Zhao, “Acoustic Inverse Rendering for Room Reconstruction and Reflectance Estimation”. (NeurIPS 2026)
- Chong Jing, **Zitong Lan**, Junan Zhang, Zhizheng Wu, “EigeNet: Geometry-Informed Multi-Modal Learning for Few-shot Novel View RIR Prediction”. (Arxiv 2026)
- Haowen Lai, **Zitong Lan**, Dongyin Hu, Mingmin Zhao, “Surface Characterization with mmWave Signals”. (Mobisys 2026)
- Hoawen Lai, **Zitong Lan**, Mingmin Zhao, “None-Line-of-Sight 3D reconstruction with Radar”. (NeurIPS 2025)([Project Site](#))
- Jinlian Hu, Yuwen Yin, **Zitong Lan**, Fusang Zhang et. al., “Quantum Sensing for Fine-Grained Liquid Recognition Using Rydberg Atoms”. (IEEE Comm. Magazine 2026)
- Fusang Zhang, Beihong Jin, **Zitong Lan**, Zhaoxin Chang, Daqing Zhang, Yuechun Jiao, Meng Shi, Jie Xiong. “Quantum Wireless Sensing: Principle, Design and Implementation”. (Mobicom 2023)
- Tengxiang Zhang, **Zitong Lan**, Chenren Xu, Yanrong Li, Yiqiang Chen. “BLEselect: Gestural IoT Device Selection via Bluetooth Angle of Arrival Estimation from Smart Glasses”. (UbiComp 2023)
- Yuechun Jiao, Jinlian Hu, **Zitong Lan**, Fusang Zhang, Jie Xiong et. al., “Exploring quantum sensing for fine-grained liquid recognition”. (Arxiv 2024)
- Zhenhao Ji, Yu Tian, Jifu Wang, Mingyuan Ding, Haoxin Wang, Yifan Chen, Jiahao Wen, **Zitong Lan**, Huiting Xu et. al. “PCCR Based Wheelchair Control System”. (IEEE Circuits and Systems Magazine 2021)

RESEARCH EXPERIENCE

- **Research Assistant, University of Pennsylvania** *Sept. 2023 - Now*
Topics: Multimodal-learning, Generative Model.
Supervisor: Prof. Mingmin Zhao
- **Research Intern, Institute of Software** *Aug. 2022 - May. 2023*
Topic: Quantum wireless sensing with Rydberg atoms.
Supervisor: Prof. Fusang Zhang and Jie Xiong
- **Research Intern, University of California Los Angeles** *Oct. 2022 - Jan. 2023*
Topic: Intermittent computing on IoT devices.
Supervisor: Prof. Yang Zhang
- **Research Intern, Institute of Computing Technology** *May. 2021 - Aug. 2022*
Topic: Bluetooth Low Energy for human interaction.
Supervisor: Prof. Tengxiang Zhang

HONORS AND AWARDS

- CVPR'25 and NeurIPS'25 distinguished reviewer *Mar. 2025*
- Howard Broadwell Fellow from Upenn *Mar. 2023*
- The Final Winner (1st) in the 2019-2020 IEEE CASS Student Design Competition *Sep. 2020*
- The Southeast University President Scholarship (university wide 2%) *Fall 2020*
- The Second Award in the Chinese Mathematics Competition *Nov. 2020*

SERVICES

Conference Reviewer: NeurIPS (2025, 2026), ICLR (2026, 2027), ICML 2026, AAAI 2027, CVPR 2025, Siggraph Asia, BMVC 2026
Journal Reviewer: TVCG, TASLP

SKILLS SUMMARY

- **Programming languages:** Python, Matlab, C, C++, Verilog
- **Software:** Audacity, Blender, Gnu-Radio, Vivado